Suggest, Don't Decide: A Dictionary-First Romanizer for Khmer Personal Names and a Leakage-Aware Evaluation

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Abstract

Khmer personal names have no single Latin spelling: តាំង is officially *Taing* for some families and *Tang* for others, and only the name owner knows which. We present khmer-name-romanizer, an open-source, zero-dependency TypeScript library that returns a ranked list of candidate spellings rather than a single answer. It combines a learned token dictionary, dictionary-guided segmentation, a rule engine with spelling conventions confirmed by a native reader, and learned spelling alternations, with no third-party model or library. It ranks full-name candidates by summed log-probability. We also report an evaluation audit: splitting by row instead of by person, scoring the reviewer's own rows, and folding over held-out data inflated accuracy on unseen names from 24.4% to 29.6% for the same code. Under the corrected protocol the system reaches 31.5% exact and 45.4% top-5 on unseen names (5-fold cross-validation), against 16.5% exact for plain grapheme rules, and 31.8% exact on a lockbox read once per release.

1 Introduction

Cambodian schools, employers, banks, and registries keep the same person in two scripts. Khmer is an abugida with stacked subscript consonants and vowel signs written around the base, so letter-by-letter conversion fails. Scholarly schemes such as ALA-LC [5] give diacritic-heavy output that no passport uses. The practical convention is a loose, mostly French-influenced spelling chosen at registration, and it is one-to-many: ប៉ែន is *Penn* (Penn Nouth) and *Pan* (Pan Ron). A single-output transliterator [3, 4] is therefore wrong by construction for part of the population.

We treat the task as *suggestion*: return ranked candidates for a person or a registrar to confirm, and a match score for search and know-your-customer (KYC) checks. Our contributions are: (1) an offline, dependency-free engine with calibrated candidate ranking, which matters because names are personal data; (2) a leakage-aware protocol for small name corpora, with an audit that measures how much a naive protocol overstates accuracy; and (3) a measured method to turn a native reader's spelling knowledge into rules, which accepts a rule only when it helps on held-out folds.

2 System

Pre-processing. The engine removes zero-width characters and leading honorifics (e.g. សម្តេច, លោក, ឯកឧត្តម). It reports the honorifics separately and splits the name on spaces.

Token resolution. Each token goes through three tiers, and the first tier that knows the token gives the leading spelling. (a) A *token dictionary* of 1,617 Khmer→Latin entries with a canonical spelling and attested variants, mined from aligned bilingual pairs. A spelling needs two independent attestations, or one when it fills a gap and agrees with the rule reading. A native reader reviews the canonicals. (b) *Segmentation*: a longest-match dynamic program splits unspaced tokens (ចាន់ដារា → Chan Dara). It is trusted only when every unknown remainder has at least three characters. (c) A grapheme-level *rule engine* (below).

Name conventions. A plain phonetic reading gives *Saret*, *Kaev*, *Lieng*, *Chai*, *Tang*, but registries write *Sareth*, *Keo*, *Leang*, *Chhay*, *Taing*. In five rounds, we showed a native reader the most frequent residual edits on the development split and asked for the convention behind them. Each proposed convention was tested alone on paired cross-validation and kept only if it raised exact or top-5 matches without losing famous names. In the fifth round, candidates came from three of the five folds, and a convention was kept only if it also helped on the other two. 33 became default readings (e.g. final dental → *th*, ម៉ៅ → *Mao*, ឆៃ → *Chhay*, តាំង → *Taing*, and *o* or *u* by consonant series: ខុន *Khon*, ហ៊ុន *Hun*); four with no measurable effect were kept on the reader's word. Six that the reader endorsed were rejected because they lowered accuracy, e.g. reading ចិន as *Chen* lost 15 rows (Chin, Sin, Rith). The plain reading stays as an equal-weight alternative, because both spellings are in use.

Learned alternations. For tokens that only the rules can read, we learn 457 context-dependent rewrites (left context, from, to, right context, probability) by aligning the plain rule reading with the official spelling, e.g. *Lav*→*Lao*. Up to two rewrites per token join the candidates, ranked below the rule reading. Promoting likely rewrites to the first guess gave +12/−8 rows (sign test p=0.5), so we did not ship it.

Ranking. Each token also collects dictionary variants and attested alternations (*nh/gn*, *n/nn*, *i/y*). A full-name candidate scores $\sum_i \log P(s_i \mid k_i)$, where the masses come from attestation counts, rule readings act as a prior, and unattested spellings get a small floor. Corpus counts cannot demote reviewed canonicals: allowing it lowered famous-name accuracy from 70/72 to 52/72. An order-flip mode handles Khmer's surname-first order for matching.

3 Data and Evaluation Protocol

The corpus has 2,898 Khmer–Latin name pairs from Wikidata/Wikipedia [6], openly licensed name lists, and distilled name lists. Foreign names written in Khmer (1,590) are out of scope. We score in-domain rows (Cambodian citizens, curated name lists, historical/regnal names) after hygiene filtering: 692 development rows and a 129-row frozen *lockbox*. The protocol has four rules:

- **Person-grouped splits.** Rows that share a normalized Khmer name, a Wikidata ID, or a multi-word Latin name form one group, and splits never divide a group [1].
- **Counterfactual training.** In each fold, a dictionary entry survives only if the training rows alone would have admitted it, and the alternation table is re-learned from the training rows.
- **No self-scoring.** Pairs written by the native reviewer are training-only, because the same person pins the canonicals.
- **Lockbox.** The lockbox is never mined or tuned on, and we

System (unseen names)	Exact	Top-5	Sim.
5-fold CV, dev split (n=692)			
Plain rules, single reading	16.5	–	.699
Rules with name conventions	25.6	34.1	.742
+ dictionary, segm., ranking	**31.5**	42.6	**.764**
+ learned alternations (v0.6.0)	**31.5**	**45.4**	**.764**
v0.5.0	29.5	43.4	.755
v0.4.0 (no conventions/alternations)	25.6	34.1	.743
Lockbox, read once per release (n=129)			
v0.4.0	24.8	31.0	.707
v0.5.0	28.7	38.8	.715
v0.6.0	**31.8**	**40.3**	**.725**
In-sample (names in the dictionary)			
Famous Cambodians (n=72)	97.2	97.2	–
Shipped dictionary, dev rows	68.1	75.7	.874

Table 1: Accuracy (%). 95% CIs (v0.6.0): CV exact 28.2–35.1, CV top-5 41.7–49.1, lockbox exact 24.4–40.2. Rule rows use no dictionary. Conventions were selected on dev CV, so CV gains for them are optimistic; the lockbox was not used for selection.

Same code, exact (%)	Naive (v1)	Corrected (v2)
5-fold CV	29.6	24.4
Lockbox	25.7	24.0

Table 2: Leakage audit on pre-release code. v1 hashed single rows into splits, folded round-robin over sorted rows (placing near-duplicates in different folds), scored reviewer-written pairs, and ran CV over the lockbox.

read it once per release.

Metrics are exact match, top-5 recall, match@10 (order-insensitive), and mean similarity (1− normalized Levenshtein distance), with 95% Wilson intervals [7].

4 Results

Main results (Table 1). The name conventions are the largest single gain: +8.5 points exact as a single reading (16.5→25.0%). The dictionary, segmentation, and ranking add 5.9 points exact over the rules, and the learned alternations add 2.8 points top-5 without changing the first guess. Against v0.4.0 on the same 692 rows, exact rises 25.6→31.5% and match@10 rises 35.1→46.8% (CI 43.1–50.5). The fifth convention round alone gives exact +15/−1 rows (sign test p=0.0005) and top-5 +18/−4 (p=0.004) against v0.5.0, and +5/−1 and +8/−2 on the two confirmation folds. The lockbox, which no selection step saw, confirms the gains (24.8→28.7→31.8% exact, 31.0→38.8→40.3% top-5). By category, CV exact is 32.2% for citizens (n=397, top-5 49.1%), 66.1% for curated lists (n=109), and 9.7% for historical/regnal names (n=186).

Leakage audit (Table 2). Before release, an audit found four leaks in the "leakage-free" protocol: two spellings of one person on both sides of a split, the reviewer's own pairs scored against the reviewer's pins, dictionary keys whose only evidence was a lockbox row, and CV folds that contained lockbox rows. Fixing them lowered CV exact by 5.2 points for the same code. Of seven lockbox "gains" from manual correction rounds, about two were genuine. Small name corpora contain many near-duplicates, so such leaks are easy to make [2].

Where the errors are. Of the 378 CV rows whose correct spelling is not in the top-5, 351 (93%) contain a token that the dictionary does not know. Only 14 have the correct option generated but ranked out. The conventions and learned alternations moved 87 rows out of this class (438→351), so better readings for unknown tokens still pay. Other ideas did not: a learned syllable compositor lost to the rules as a single guess (72 vs. 76 exact on 248 leave-one-out tokens), segmentation changes rescued at most 3 of 102 misses, and three v0.6.0 probes were rejected: reranking a larger candidate pool with a letter model (top-5 +17/−9, p≈0.2), extra spellings for rarely seen dictionary words (+4/−1), and spacing compound given names (top-5 cannot gain; exact fell). Ranking is not the bottleneck: only 24 missed rows have the official spelling among the generated options at all.

5 Discussion and Limitations

Misses are mostly near misses (mean similarity 0.76, e.g. *Penn Rorn* for *Pan Ron*). Two limits remain that algorithms cannot remove: the correct spelling is a family's decision at registration, and public sources have limited vocabulary. The conventions come from one native reader, and their selection on dev CV makes CV slightly optimistic; the lockbox interval is wide (n=129). Variants merged into dictionary entries can still leak, so top-5 is slightly optimistic. The corpus over-represents public figures. Because the engine cannot know an official spelling, the library is a pre-fill for human confirmation and must not write spellings onto identity documents automatically. It runs fully offline, so names never leave the host process.

Availability. MIT license. npm: `khmer-name-romanizer` v0.6.0. Code, corpus tools, and evaluation scripts: `https://github.com/hangsopheak/khmer-name-romanizer`.

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